

Design History File (DHF)

Wearable ECG Patch

Automated DHF Project - Shivangi Sarkar

Device Overview

Wearable ECG Patch

Class II medical device

Purpose: Wearable ECG patch is a diagnostic tool which helps monitor heart rhythm and can prevent fatal conditions like Atrial Fibrillation and reduce stroke risk.

Features:

- Electrodes
- Analog front end
- Microcontroller unit
- Bluetooth Low Energy Transceiver for wireless transmission
- Power supply
- Software system
- Mobile app
- PC Interface
- Posture recognition module

End Users: Cardiac Patients and Clinicians who are monitoring the patch

Environment: Outside of hospital

Regulatory Class: FDA Class II

User Needs

Need ID	Description
UN001	must accurately detect cardiac events
UN002	must be able to monitor long term
UN003	must be able to function remotely
UN004	must be comfortable to wear
UN005	must be easy to setup app as well as the device
UN006	must have minimal user interaction after setup
UN007	must be discreet for social setting
UN008	must be compatible with skin (skin irritation, allergic reaction)
UN009	must be wire free for better mobility and comfort
UN010	must be non-intrusive (not interfere with daily lifestyle)
UN011	must be easily replaceable or maintained

Design Inputs

Input ID	Description	Linked Need ID
DI001	The device shall use a medical-grade ECG front-end (e.g., ADAS1000) with a sampling rate $\geq$ 250 Hz and signal-to-noise ratio $\geq$ 20 dB.	UN001
DI002	The device shall store and/or transmit ECG data continuously for $\geq$ 7 days with 1-minute resolution.	UN002
DI003	The device shall support Bluetooth Low Energy (BLE 5.0) with automatic data syncing to cloud via mobile app.	UN003
DI004	The patch shall weigh $<$ 25g and use breathable, hypoallergenic adhesive suitable for continuous wear.	UN004
DI005	The mobile app shall feature an onboarding wizard and video instructions for setup.	UN005
DI006	The device shall automatically start monitoring after setup and require no further input during regular operation.	UN006
DI007	The patch shall have a form factor $<$ 5mm thick with skin-tone or transparent housing for discreetness.	UN007
DI008	The adhesive and outer casing shall be ISO 10993 biocompatible and dermatologically tested.	UN008
DI009	The patch shall operate wirelessly with internal battery and zero wired components during use.	UN009
DI010	The device shall allow full range of motion (e.g., sitting, walking, sleeping) without mechanical interference.	UN010
DI011	The device battery and adhesive shall be modularly replaceable by user in $<$ 1 minute using a slide-lock mechanism.	UN011

Design Outputs

Output ID	Description	Linked Input ID
DO001	ECG signal acquisition validated using ADAS1000 at 250 Hz in test bench and signal-to-noise ratio confirmed ≥ 20 dB.	DI001
DO002	Data logging and transmission verified with 7-day continuous operation in simulated patient environment.	DI002
DO003	BLE 5.0 module integrated with successful data transmission test to Android/iOS app; encryption logs verified.	DI003
DO004	Device weight measured at 22g; adhesive passed 7-day continuous wear comfort test.	DI004
DO005	Mobile app onboarding tested with 10 users; setup completed in < 3 minutes with 95% satisfaction rate.	DI005
DO006	Software confirmed to auto-start monitoring post-setup without manual intervention during 72-hour test cycle.	DI006
DO007	Final patch enclosure is 4.5mm thick with neutral tone; confirmed unnoticeable under clothing in user trial.	DI007
DO008	ISO 10993 biocompatibility reports completed and passed for adhesive and enclosure; no irritation in 10-user patch test.	DI008
DO009	Wireless functionality tested under movement and daily activities; no disconnection over 72-hour dynamic test.	DI009
DO010	Patch tested on users during sitting, walking, and sleeping; motion artifact < 5% in ECG recording.	DI010
DO011	Battery and adhesive module replaced by users in < 1 minute; validated through usability testing.	DI011

Verification Methods

Input ID	Verification Method
DI001	Bench test ECG signal using waveform simulator and confirm sampling rate and signal-to-noise ratio meet specs.
DI002	Continuous data logging test in simulated use case for 7 days with battery life monitoring and data integrity check.
DI003	Perform Bluetooth Low Energy (BLE) range test and validate AES encryption via transmission log analysis.
DI004	Measure device weight and conduct human wear test for 7 days with comfort feedback survey.
DI005	Usability test with 10 users setting up device and app; measure completion time and error rate.
DI006	Run 72-hour software test cycle to ensure automatic monitoring starts post-setup with no manual actions.
DI007	Measure patch thickness and conduct visibility test under clothing on a diverse user group.
DI008	Submit adhesive and enclosure for ISO 10993 biocompatibility testing and patch wear irritation test.
DI009	Wireless performance test with daily activity simulation and real-time disconnection logging.
DI010	Motion artifact evaluation during sitting, walking, and sleeping using annotated ECG signal recordings.
DI011	Conduct usability test where users replace battery and adhesive within 1 minute; measure completion time and error rate.

Change Log

Date	Change Description	Author
2024-06-01	Initial design inputs uploaded	Shivangi
2024-06-10	Updated risk assessment thresholds	QA Engineer
2024-06-15	Added verification test cases	Design Lead

Traceability Matrix

Need_ID	User_Need_Description	Input_ID	Input_Description	Output_ID	Output_Description	Verification_Method
UN001	must accurately detect cardiac events	DI001	The device shall use a medical-grade ECG front-end (e.g., ADAS1000) with a sampling rate ≥ 250 Hz and signal-to-noise ratio ≥ 20 dB.	DO001	ECG signal acquisition validated using ADAS1000 at 250 Hz in test bench and signal-to-noise ratio confirmed ≥ 20 dB.	Bench test ECG signal using waveform simulator and confirm sampling rate and signal-to-noise ratio meet specs.
UN002	must be able to monitor long term	DI002	The device shall store and/or transmit ECG data continuously for ≥ 7 days with 1-minute resolution.	DO002	Data logging and transmission verified with 7-day continuous operation in simulated patient environment.	Continuous data logging test in simulated use case for 7 days with battery life monitoring and data integrity check.
UN003	must be able to function remotely	DI003	The device shall support Bluetooth Low Energy (BLE 5.0) with automatic data syncing to cloud via mobile app.	DO003	BLE 5.0 module integrated with successful data transmission test to Android/iOS app; encryption logs verified.	Perform Bluetooth Low Energy (BLE) range test and validate AES encryption via transmission log analysis.
UN004	must be comfortable to wear	DI004	The patch shall weigh < 25 g and use breathable, hypoallergenic adhesive suitable for continuous wear.	DO004	Device weight measured at 22g; adhesive passed 7-day continuous wear comfort test.	Measure device weight and conduct human wear test for 7 days with comfort feedback survey.
UN005	must be easy to setup app as well as the device	DI005	The mobile app shall feature an onboarding wizard and video instructions for setup.	DO005	Mobile app onboarding tested with 10 users; setup completed in < 3 minutes with 95% satisfaction rate.	Usability test with 10 users setting up device and app; measure completion time and error rate.
UN006	must have minimal user interaction after setup	DI006	The device shall automatically start monitoring after setup and require no further input during regular operation.	DO006	Software confirmed to auto-start monitoring post-setup without manual intervention during 72-hour test cycle.	Run 72-hour software test cycle to ensure automatic monitoring starts post-setup with no manual actions.

UN007	must be discreet for social setting	DI007	The patch shall have a form factor < 5mm thick with skin-tone or transparent housing for discreetness.	DO007	Final patch enclosure is 4.5mm thick with neutral tone; confirmed unnoticeable under clothing in user trial.	Measure patch thickness and conduct visibility test under clothing on a diverse user group.
UN008	must be compatible with skin (skin irritation, allergic reaction)	DI008	The adhesive and outer casing shall be ISO 10993 biocompatible and dermatologically tested.	DO008	ISO 10993 biocompatibility reports completed and passed for adhesive and enclosure; no irritation in 10-user patch test.	Submit adhesive and enclosure for ISO 10993 biocompatibility testing and patch wear irritation test.
UN009	must be wire free for better mobility and comfort	DI009	The patch shall operate wirelessly with internal battery and zero wired components during use.	DO009	Wireless functionality tested under movement and daily activities; no disconnection over 72-hour dynamic test.	Wireless performance test with daily activity simulation and real-time disconnection logging.
UN010	must be non-intrusive (not interfere with daily lifestyle)	DI010	The device shall allow full range of motion (e.g., sitting, walking, sleeping) without mechanical interference.	DO010	Patch tested on users during sitting, walking, and sleeping; motion artifact < 5% in ECG recording.	Motion artifact evaluation during sitting, walking, and sleeping using annotated ECG signal recordings.
UN011	must be easily replaceable or maintained	DI011	The device battery and adhesive shall be modularly replaceable by user in < 1 minute using a slide-lock mechanism.	DO011	Battery and adhesive module replaced by users in < 1 minute; validated through usability testing.	Conduct usability test where users replace battery and adhesive within 1 minute; measure completion time and error rate.

Risk Management

Failure Mode	Effect	Cause	Severity (S)	Occurrence (O)	Detection (D)	RPN	Recommended Action
ECG signal noise	Misdiagnosis	Loose electrode connection	9	4	3	108	Improve electrode placement & shielding
BLE transmission fails	Data loss	Interference, low battery	8	3	4	96	Add auto-reconnect and alert system
Device adhesive detaches	Interrupted monitoring	Sweat, poor skin contact	7	4	4	112	Redesign adhesive for better grip
Skin irritation	User discomfort	Poor adhesive material	5	3	5	75	Switch to ISO 10993-certified adhesive
Setup errors	Device not collecting data	Poor user onboarding design	8	2	3	48	Simplify app UI, add guided tutorial
Patch visibility concern	Non-compliance with use	Bulky design	4	3	3	36	Reduce thickness and use neutral tones
API sync failure	Missed real-time alerts	Backend or app issue	7	3	4	84	Add sync verification + alert retry
Battery dies early	Device shutdown	High power drain	9	2	3	54	Optimize power consumption logic
Lead-off not detected	Inaccurate ECG reading	Software bug	10	2	3	60	Validate detection logic extensively

